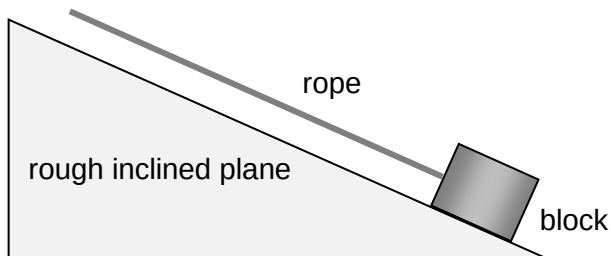


- 11** A block is being pulled up a rough inclined plane using a rope at a constant speed.



Which of the following statements is correct?

- A** The weight of the block does no work because the block is not moving in the direction of its weight.
- B** The work done by the force of the rope is equal to the work done against the frictional force from the plane because there is no gain in kinetic energy.
- C** The normal reaction force from the plane does positive work because it has an upward component and the block is moving upwards.

- D** The negative work done by the weight of the block is equal to the gain in gravitational potential energy of the block.

Ans: D

A force does no work when it is perpendicular to the direction of motion. (A & C)

Net work done on the block is positive because there is gain in GPE of the block. (B)

- 12** The figure below shows a light inextensible rope that passes over a light smooth pulley with two